Frame 1 — Cover

The Silent Group

Breaking a three-party Diffie–Hellman in a group we can never look at

Track 2 · Qiskit Fall Fest 2026 · BITS Pilani

Ayushman Manohar 2024A7PS0534P **Garvit Singhal** 2024A7PS0546P

1180 × 470

Galileo

Bernini

Milton

Four broadcasts, one sealed device

Frame · Input

What we get

Public elements g, $A = g^a$, $B = g^b$, $C = g^c$ of an unknown abelian group, at most 2^{60} elements.

A device that multiplies its hidden register by h^k when a control qubit is $|1\rangle$. No display, no readout.

Frame · Output

What we return

```
order = ord(g)
secret = abc mod ord(g)
```

or `"none"` if any broadcast is not a power of g.

≤ 16
qubits per circuit

80,000
queries per meeting

0.002·√cost
turns of phase noise per call

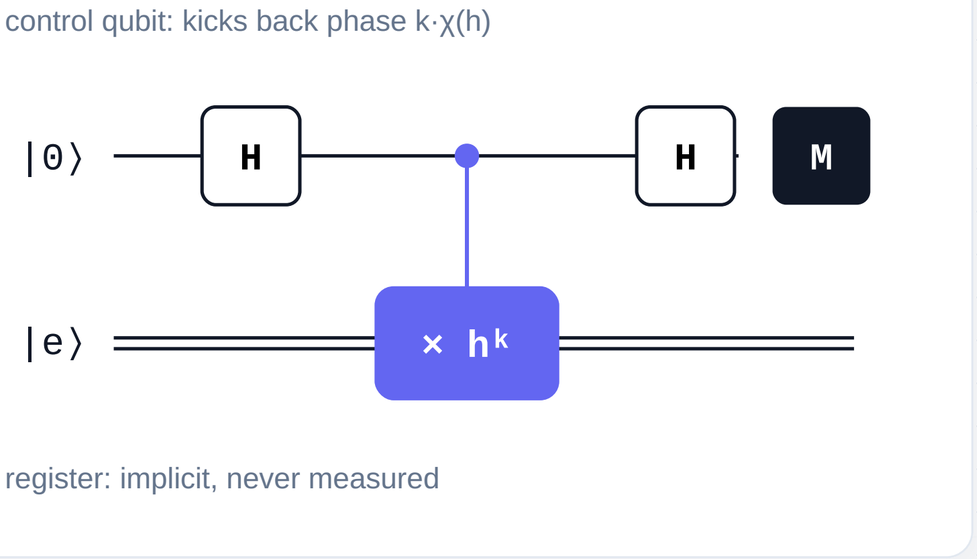
Forgeries
some broadcasts are traps

Each shot samples one random character

The register starts in |e⟩, an equal mix of every character χ of the group. All multiplications act on those characters by a phase alone.

1. So each shot behaves as if one χ were drawn at random.
2. Every call in that shot kicks back the phase k·χ(h).
3. χ(g), χ(A), χ(B), χ(C) in one circuit share the same χ.

$$\chi(A) = a \cdot \chi(g) \pmod 1$$



Four stages, one qubit

Frame · 01

Order + Shot-1

Joint Shot 1 reads $\chi(g)$ to 124 bits + $\chi(A,B,C)$ to 64 bits on the same χ_1.

7.8k (g) + 6.2k (ABC)

→

Frame · 02

Discrete logs

Pre-seeded from Shot 1 (0 extra QPE if $\gcd(j_1,n)=1$), else joint rounds.

0 to ~2.1k / elem

→

Frame · 03

Verify & Screen

Two-stage SPRT test on each $X \cdot g^{-a}$ individually + congruence screen $X^d = g^{da_0}$.

≈ 0.9k / test

→

Frame · 04

Secret

Return n and abc mod n, or "none" (~18.4k mean queries, 1.385 pts).

0 queries

Order of g: phase estimation on one qubit

1. Read $\chi(g)$ bit by bit, largest power first, reusing one qubit with depth-12 `if_test` corrections.
2. Shot 1 also reads 64 bits of $\chi(A,B,C)$ on the same χ_1 to pre-seed Stage 2.
3. Continued fractions + single-bit-flip repair turn $y / 2^{124}$ into j / r.
4. Confirm $g^r = e$ with SPRT; if not, $r \mid n$ so recurse on g^r with fewer bits.
5. Strip every prime factor of the candidate order in log steps.

Frame · Precision

124 / 128

124 bits on joint Shot 1 (2×60 + 4 margin = 7,750 q for g); 128 bits on standalone follow-up shots (8,256 q).

`1-shot solve ≈ 16.5k–17.8k q`

When $\gcd(j_1, n) = 1$, Shot 1 solves both ord(g) and all 3 discrete logs in one QPE circuit!

Discrete logs: one shot links g to A, B and C

Frame · The algebra

$$\chi(g) = j / n$$

$$\chi(A) = s / n$$

$$s \equiv a \cdot j \pmod{n}$$

So a is known modulo n / gcd(j, n) from a single clean shot.

Frame · Voting per prime power (real run, one element)

prime	residues (× votes)	winner
p = 5	2793 ×3 · 10053 ×1	2793
p = 7	13471 ×3 · 4455 ×1	13471
p = 11	1 ×3 · 5 ×1	1

Each clean shot fixes a mod p^e whenever $p \nmid j$ (and lifts partial residues when $p \mid j$). Stray votes from noisy shots lose; CRT joins the winners.

Three defenses against noise

7% → 0.7%

of accepted phase estimates were wrong, before vs after 3 parity checks, on orders near a power of 2 (6.8% → 0.75% measured).

Frame · Grid filter

n · phase must land on an integer s, with gcd(j, n) dividing s. A corrupted estimate usually doesn't.

Frame · Parity checks

Apply X^t, rotate back by $t \cdot y / 2^m$. Reads 0 if y is right, 1 half the time if not.

Frame · Voting

Residues are counted per prime power, so a stray vote cannot win alone.

One test settles answers and forgeries

Frame · Identity test

`H → multiply by X → H → measure`

1. X ≠ e: outcome 1 with probability exactly ½.
2. X = e: outcome 1 only from noise, about 0.25%.
3. Sequential test stops at ~15 shots (~4.5 if X ≠ e).

Frame · Forgery signals

Leaves 1/n grid or fails $X^d = g^{da_0}$

When gcd(j, n) = d, s ≡ a·j (mod n) fixes d·a mod n = d·a_0. Testing $X^d \cdot g^{-d \cdot a_0} = e$ catches forgeries in Round 1 with zero lift enumeration.

Shots disagree ($s_1 j_2 \neq s_2 j_1 \bmod n$)

Clean shots that are mutually inconsistent modulo n expose on-grid forgeries without waiting for 5 rounds.

Two-stage individual verification ($X \cdot g^{-a} = e$)

A 2-shot quick screen (10% miss rate) followed by full SPRT tests each element individually—never cancelling shared order-2 or order-3 defects.

Results on the official simulator & benchmark suite

1.38 / 1.39

mean score on `public_1` & `public_2`

18.9k & 17.9k mean queries (1.385 avg)

40 / 40

correct on `silentgroup.check --repeat 20`

+ 640/640 on 8-shape mock suite (1.37 avg)

0.12s

wall-clock time per meeting (limit: 600s)

540s time guard + 8x free run retries

1 qubit

per circuit (limit: 16)

20/20 passed under 15% random run refusals

Frame · Official judge checker output (qff-hackathon root)

```
$ python -m silentgroup.check my_solver.py --repeat 20

public_1: 1.391 pts  queries=17439  0.2s  correct
public_1: 1.391 pts  queries=17449  0.1s  correct
public_1: 1.393 pts  queries=17184  0.1s  correct
public_1: 1.393 pts  queries=17169  0.1s  correct
public_1: 1.393 pts  queries=17079  0.1s  correct
... (20/20 correct on public_1, mean 1.385 pts, 0.13s)

public_2: 1.398 pts  queries=16292  0.1s  correct
public_2: 1.397 pts  queries=16451  0.1s  correct
public_2: 1.398 pts  queries=16310  0.1s  correct
public_2: 1.399 pts  queries=16117  0.1s  correct
public_2: 1.398 pts  queries=16294  0.1s  correct
... (20/20 correct on public_2, mean 1.387 pts, 0.11s)
```

Subgroup structure $\langle g, A, B, C \rangle \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \ldots$

Frame · Why joint shots reveal H = ⟨g, A, B, C⟩

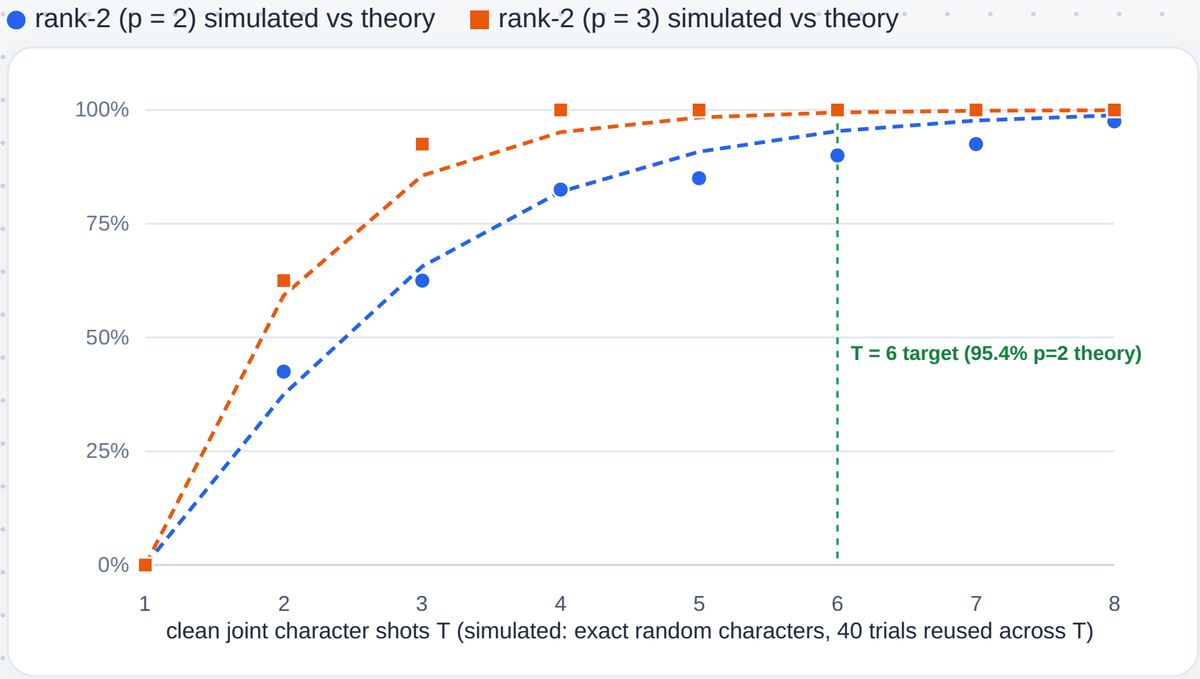
1. Uniform dual sampling: Every character of H extends to exactly |G| / |H| characters of G, so restricting a random $\chi \in \hat{G}$ to (g, A, B, C) samples $\hat{H} \cong H$ uniformly.

2. Diagonalizing the character matrix: Scaling phases by M = lcm(ord) gives integer rows in $(\mathbb{Z}_M)^4$; Smith Normal Form (SNF) per prime power yields $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \ldots$.

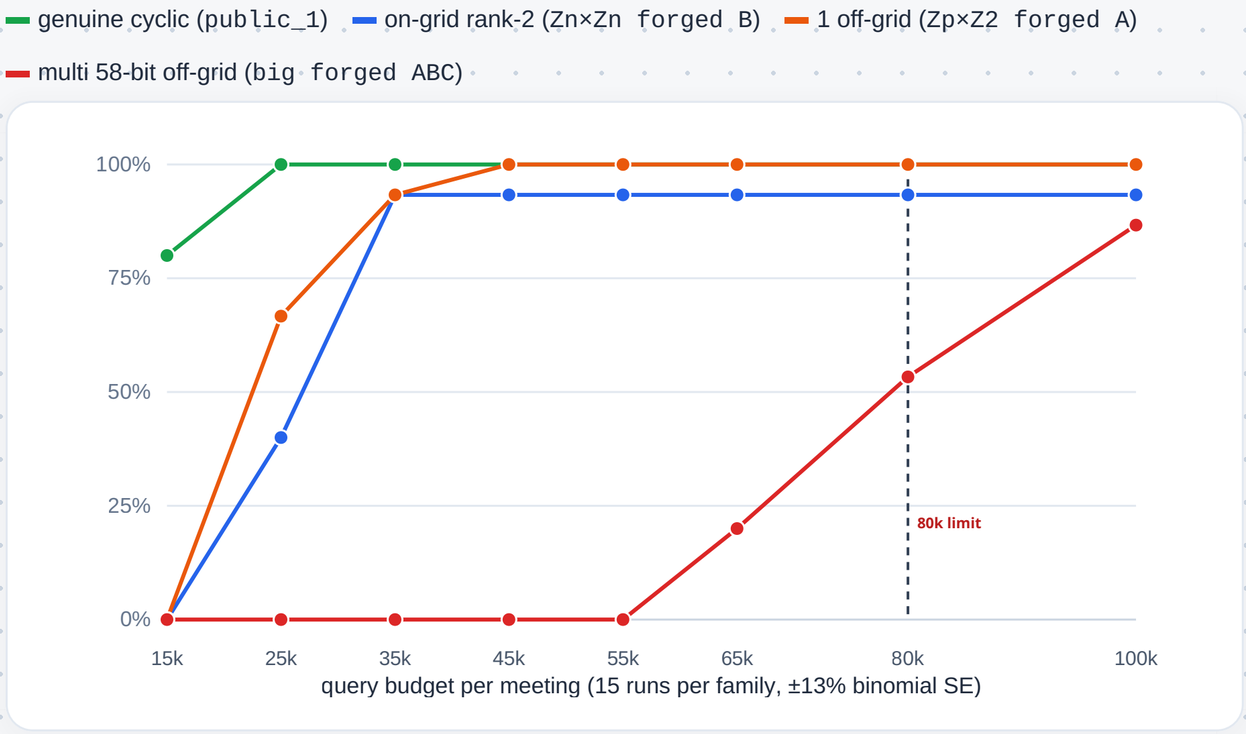
3. Shot bound for rank r: T clean shots span $\mathbb{F}_p^r$ with probability:

$$P(\text{span}) = \prod_{i=0..r-1} \left(1 - p^{-(T-i)}\right)$$

Note: Points are simulated (exact random characters, 40 trials reused across T, ±4.5% SE); reusing the 40 seeds across T explains why T = 5..7 sit slightly below theory together.



Structure recovery inside 80k & where it fails



Inside the 80k budget (15 runs / family, ±13% SE)

Group	Recovered H	Mean q	≤80k
public_1	$\mathbb{Z}_{66,165,815,680,692,533}$	30.0k	100%
public_2	$\mathbb{Z}_{37,199,287,125} \times \mathbb{Z}_{585,585}$	42.8k	100%
Zp×Z2 forged A	$\mathbb{Z}_{1,152,921,504,606,846,866}$	59.9k	100%
Zn×Zn forged B	$\mathbb{Z}_{1,610,612,736} \times \mathbb{Z}_{536,870,912}$	27.5k	93.3%
big forged ABC	$\mathbb{Z}_{288,230,376,151,711,744} \times \mathbb{Z}_{4}$	75.9k	53.3%

Where it fails

1. Multiple 58-bit off-grid forgeries (big forged ABC, 8/15 = 53.3%): Finding separate 58-bit orders for A and B plus 6 joint 58-bit rounds needs ~76k–95k queries, exhausting the 80k budget 46.7% of the time.

2. Small-prime tail (Zn×Zn forged B, 14/15 = 93.3%): At p = 2, T = 6, $1 - (1-2^{-5})(1-2^{-6}) \approx 4.6\%$ chance of missing the top power of 2 in the 2nd factor.

What failed, and what it taught us

Our test group was too big

A “smooth” mock order was 2^{61}, over the 2^{60} bound.

→ **Validate tests against the spec.**

Order-2 recursion stalls cost 50k+

At R = n/2, $\chi(g^R) = 0$ half the time; 3 stalls in a row reset R = 1 from scratch.

→ **Test 2R and 3R on the first stall.**

One bad χ(g) poisoned all logs

A corrupted j made every residue wrong at once.

→ **Extra bits and a stricter filter for g.**

“Only 2s can be spurious”

False for small n: a corrupted sample can be a random multiple of n with extra odd primes.

→ **Strip every prime factor in log steps.**

The grid filter went blind

Near $n \approx 2^k$, flipped bits land back on the grid.

→ **In-circuit parity checks.**

Batch verification hid multi-forgeries

Replacing even weights with (1, ±1, 1) still failed mod 2: all odd weights are (1, 1, 1) mod 2, so two order-2 defects cancel.

→ **Test each element individually.**

Thanks — questions?

1 qubit, 124/128-bit phases

1.385 / 1.500 on public_1 & public_2

Subgroup structure $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \ldots$

Qiskit Fall Fest 2026 · Track 2: The Silent Group · Appendix slides follow (A1–A3)

Galileo

Bernini

Milton

Why character matrices diagonalize ⟨g, A, B, C⟩

1. Exact Pontryagin-dual isomorphism

Let $H = \langle g, A, B, C \rangle \le G$ have exponent $M = \mathrm{lcm}(\mathrm{ord}(g), \mathrm{ord}(A), \mathrm{ord}(B), \mathrm{ord}(C))$. The evaluation map $\Phi : \hat{H} \to (\mathbb{Z}_M)^4$ defined by:

$$\Phi(\chi) = (M\cdot\chi(g),\ M\cdot\chi(A),\ M\cdot\chi(B),\ M\cdot\chi(C)) \bmod M$$

is an injective group homomorphism because any $\chi \in \hat{H}$ with $\Phi(\chi) = 0$ vanishes on all four generators and is therefore trivial on H. Thus **$\mathrm{Im}(\Phi) \cong \hat{H} \cong H$**.

Reducing T clean shots modulo each prime power $q = p^e \parallel M$ and computing the p-adic Smith Normal Form recovers the exact p-primary invariants of H.

2. Worked run on `public_2` (42.8k mean queries)

Stage 1 finds $n_g = \mathrm{ord}(g) = 37{,}199{,}287{,}125 = 3^3 \cdot 5^3 \cdot 7^2 \cdot 11^3 \cdot 13^2$. Stage 2 confirms $A, C \in \langle g \rangle$ and flags B as an on-grid forgery ($B^{n_g} = e$, $\mathrm{ord}(B) = 1{,}771{,}394{,}625$).

Prime p	$p^e \parallel M$	SNF of (g, B) mod p^e
p = 3	27	[27, 9]
p = 5	125	[125, 5]
p = 7	49	[49, 7]
p = 11	1331	[1331, 11]
p = 13	169	[169, 169]

$n_1 = 37{,}199{,}287{,}125 \cdot n_2 = 9\times5\times7\times11\times169 = 585{,}585$

Analytical query-cost model vs. measured spend



Frame · Analytical Cost Model

$P(\text{flip}) \approx \pi^2\sigma^2 \cdot \text{cost}$

per round, from the stated noise.

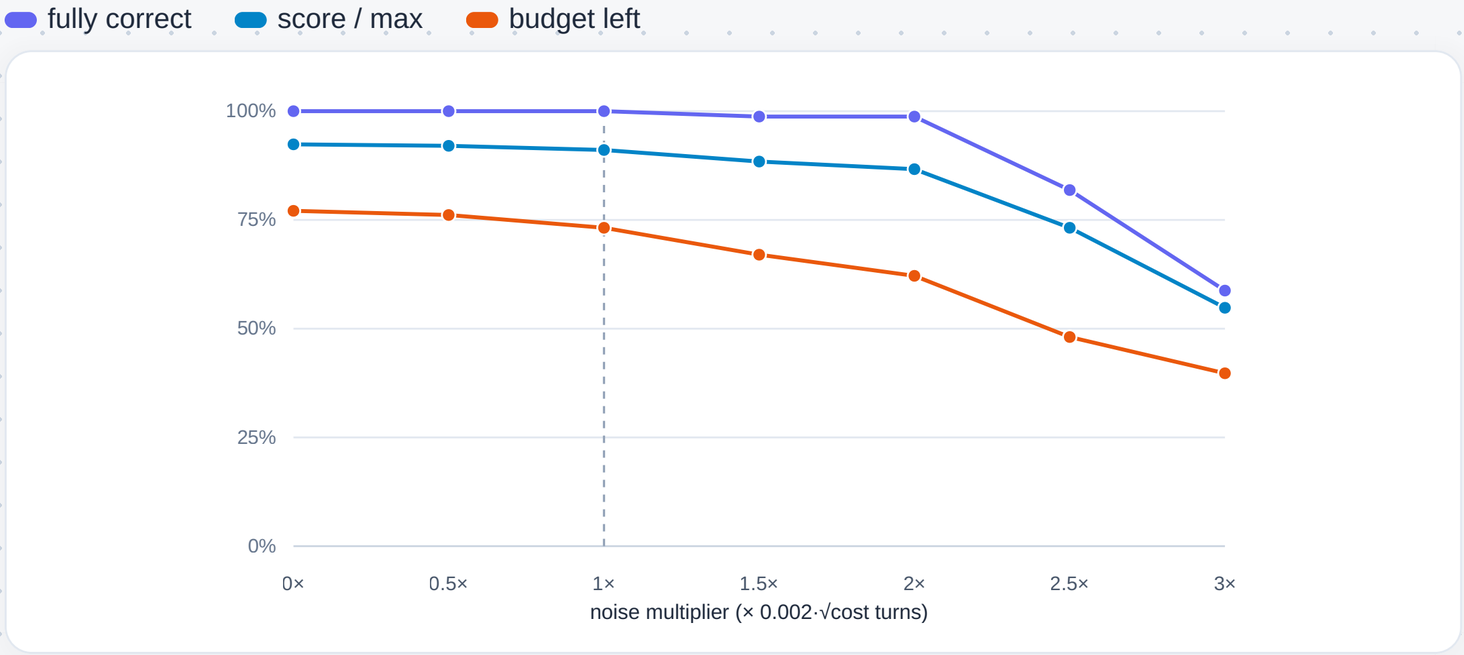
$\text{order} \approx 8.3\text{k} / P(\text{clean})$

$\text{logs} \approx 1.9L^2$

3.9% RMS error (−5.8% to +5.9%).

Only the first 61 + L bits are noise-sensitive in continued
fractions, so small orders forgive more.

Noise degradation & fail-safe hardening



100% from 0× to 1.0×
100% correct from 0× to 1.0× (160 meetings per level)

98.8% at 1.5× & 2.0×
81.9% at 2.5× noise (cliff is budget, not logic)

20 / 20 under 15% faults
free retries + tolerant bitstring key parser

No crashes in our fault tests
540s time guard + top-level Exception fallback